THE ALIGNMENT LIMIT FOR ALL PRIMES

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Abstract. The companion papers proved that the repetend alignment of $n = pm$ (with m a b-smooth integer and p a digit-partitioning prime) equals $(2m-1)/(pm-1)$. This paper gives an exact formula for all primes.

For any prime p not dividing b and any b-smooth $m \geq 1$:

$$\alpha(pm) = \frac{m - 1 + m \cdot T(C_m)/L}{pm - 1},$$

where $L = \mathrm{ord}_p(b)$, the coset $C_m = u\langle b\rangle$ is determined by $u = b^t/m$ with $m \mid b^t$, and $T(C_m) = \sum_{r \in C_m} n_{\delta(r)}$ is the coset bin sum. The formula is exact at every finite m.

The subsequential alignment limit as $m \to \infty$ depends on the coset C_m. When b is a primitive root modulo p, there is one coset and the limit is $(p-1+S)/(p(p-1))$ where $S = \sum n_d^2$. In general, different cosets yield different limits.

For all primes $p \geq 5$, all cosets, and all b-smooth m: $\alpha(pm) < 1/\varphi$. Among primes $p \geq 3$, only $p = 3$ produces alignment above the golden threshold.

1. Introduction

This paper extends the alignment formula of the preceding papers to all primes.

The first paper in this series [1] proved that the repetend alignment of $n = 3m$ equals $(2m-1)/(3m-1)$ and that the golden ratio's minimal polynomial uniquely divides the self-referential cubic at $p = 3$. The second paper [2] extended the alignment formula to all digit-partitioning primes ($p \leq b+1$) and characterized them by the injectivity of the digit function.

For primes $p > b+1$, the digit function is not injective. Fractions from different residue classes share digits at common positions, and the alignment formula $(2m-1)/(pm-1)$ underestimates the true alignment.

This paper computes the correction. We show that the alignment exact formula at every finite m is determined by the coset bin sum. The alignment never reaches $1/\varphi$ for any prime $p \geq 5$.

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2. The Digit Function and Its Bins

Definition 1. For a prime p not dividing b, the *digit function* is $\delta(r) = \lfloor br/p \rfloor$ for $r \in \{0, \ldots, p-1\}$. The *bin* of digit d is $B_d = \{r \in \{1, \ldots, p-1\} : \delta(r) = d\}$, and its size is $n_d = |B_d|$.

The $p-1$ nonzero remainders are distributed among b bins. Let $q = \lfloor (p-1)/b \rfloor$ and $r = (p-1) \bmod b$. Then r bins have size $q+1$ and $b-r$ bins have size q.

Lemma 2. *The sum of squared bin sizes is*

$$S(p,b) = \sum_{d=0}^{b-1} n_d^2 = bq^2 + r(2q+1).$$

Proof. There are r bins of size $q+1$ and $b-r$ bins of size q. Thus

$$S = r(q+1)^2 + (b-r)q^2 = rq^2 + 2rq + r + bq^2 - rq^2 = bq^2 + r(2q+1). \quad \square$$

Corollary 3. *When $p \leq b+1$, all bins have size at most 1, so $S(p,b) = p-1$.*

3. The Coset and Its Bin Sum

For $n = pm$ with m a b-smooth integer, choose t with $m \mid b^t$ and set $u = b^t/m$. The coset $C_m = u\langle b \rangle \bmod p$ is independent of the choice of t: if $m \mid b^{t'}$ as well, then $b^{t'}/m = (b^t/m) \cdot b^{t'-t}$, which lies in the same coset of $\langle b \rangle$. After t prefix digits, the repetend of $k/(pm)$ is the repetend of uk/p [2]. The relevant orbit is not the subgroup $\langle b \rangle$ but the coset C_m.

Definition 4. For a coset C of $\langle b \rangle$ in $(\mathbb{Z}/p\mathbb{Z})^*$, the *coset bin sum* is

$$T(C) = \sum_{r \in C} n_{\delta(r)},$$

where $n_d = |B_d|$ is the size of the d-th bin.

Different b-smooth values of m can yield different cosets and therefore different bin sums.

Remark 5. The coset C is *equidistributed* across bins if $|C \cap B_d| = L \cdot n_d/(p-1)$ for every d. When this holds, $T(C) = LS/(p-1)$ where $S = \sum_d n_d^2$. In the primitive root case ($L = p-1$), there is only one coset and equidistribution is automatic.

4. The Alignment Limit

Theorem 6. *For any prime p not dividing b and any b-smooth $m \geq 1$, the repetend alignment of $n = pm$ is*

$$\alpha(pm) = \frac{m - 1 + \dfrac{m \cdot T(C_m)}{L}}{pm - 1},$$

where $L = \mathrm{ord}_p(b)$ and $C_m = u\langle b\rangle$ is the coset determined by $u = b^t/m$ with $m \mid b^t$.

Proof. The $m - 1$ multiples of p in $\{1, \ldots, pm-1\}$ terminate, each contributing alignment 1. The remaining $(p-1)m$ values of k are non-terminating. After the prefix of length t, the repetend of $k/(pm)$ is the repetend of $(uk \bmod p)/p$, where $u = b^t/m$.

The repetend of $1/(pm)$ has length L. At the i-th position, the digit is $\delta(r_i)$ where r_i is the i-th element of the coset C_m. A non-terminating fraction $k/(pm)$ matches $1/(pm)$ at position i if and only if $uk \bmod p$ lies in the same bin $B_{\delta(r_i)}$, which contains $n_{\delta(r_i)}$ residues. Each residue class mod p accounts for m values of k. Averaging the match count over the L positions:

$$\text{aligned non-terminating} = m \cdot \frac{1}{L} \sum_{r \in C_m} n_{\delta(r)} = \frac{m \cdot T(C_m)}{L}.$$

Adding the $m - 1$ terminating fractions:

$$\alpha(pm) \cdot (pm - 1) = (m - 1) + \frac{m \cdot T(C_m)}{L}. \qquad \square$$

Remark 7. As $m \to \infty$, $\alpha(pm) \to (L+T(C_m))/(pL)$. The limit depends on the coset C_m. When b is a primitive root modulo p, there is only one coset and the limit is $(L+T)/(pL)$. When $L < p-1$, different b-smooth values of m may yield different cosets, and the limit over all smooth m does not exist in general.

Remark 8. A concrete example: $p = 53$, $b = 10$, $L = 13$. The subgroup $\langle 10\rangle$ has $T = 67$. The coset $5\langle 10\rangle$ has $T = 69$. For $m = 2^a$: $u = 5^a$, and the coset alternates with $a \bmod 4$, giving two subsequential limits 80/689 and 82/689.

Corollary 9. *When b is a primitive root modulo p ($L = p-1$, one coset), the alignment limit exists and equals*

$$\mathcal{L}(p, b) = \frac{p - 1 + S(p, b)}{p(p - 1)},$$

where $S(p, b) = bq^2 + r(2q+1)$ with $q = \lfloor (p-1)/b \rfloor$ and $r = (p-1) \bmod b$.

Proof. When $L = p-1$, there is only one coset and $T = S$ (every residue is visited). Substituting into Theorem 6 and taking $m \to \infty$: $\mathcal{L} = (L+S)/(pL) = (p-1+S)/(p(p-1))$. □

Corollary 10. *For digit-partitioning primes ($p \leq b+1$), $S = p-1$ and $\mathcal{L}(p,b) = 2/p$.*

Proof. By Corollary 3, $S = p-1$. Substituting into Corollary 9: $(p-1+p-1)/(p(p-1)) = 2/p$. □

5. Enumeration in Base Ten

p	L	S	Equidist.	$\mathcal{L}(p)$	Decimal
3	1	2	✓	2/3	0.6667
7	6	6	✓	2/7	0.2857
11	2	10	✓	2/11	0.1818
13	6	16	✓	7/39	0.1795
17	16	28	✓	11/68	0.1618
19	18	34	✓	26/171	0.1520
23	22	50	✓	36/253	0.1423
29	28	80	✓	27/203	0.1330
31	15	90	✓	4/31	0.1290
37	3	132	✓	14/111	0.1261
41	5	160	✓	5/41	0.1220
43	21	186	✓	110/903	0.1218
47	46	216	✓	130/1081	0.1203
53	13	272		80/689	0.1161
59	58	346	✓	198/1711	0.1157
61	60	366	✓	7/61	0.1148

Table 1. Alignment values for primes up to 61 in base 10. For primitive-root primes (checkmarked), the limit is unique. For $p = 53$ ($L = 13$, four cosets), the value shown is for the subgroup coset; other cosets give different limits.

6. The Golden Ratio Bound

Theorem 11. *For every prime $p \geq 5$ not dividing b, every base $b \geq 2$, and every b-smooth $m \geq 1$:*

$$\alpha(pm) < \frac{1}{\varphi}.$$

In particular, every subsequential alignment limit (over b-smooth $m \to \infty$) is below $1/\varphi$.

Proof. For any coset C, the bin sum satisfies $T(C) \leq L \cdot \max_d n_d$ (each of the L coset elements contributes at most $\max_d n_d$). Since $\max_d n_d \leq \lceil (p-1)/b \rceil$, the alignment at any b-smooth m satisfies

$$\alpha(pm) \leq \frac{m-1+mc}{pm-1} \leq \frac{1+c}{p} \leq \frac{p+1}{2p},$$

where $c = \lceil (p-1)/b \rceil$. The second inequality holds because $(m-1+mc)(p) \leq (1+c)(pm-1)$ simplifies to $p-1-c \geq 0$, which is true since $c \leq (p-1)/b \leq p-1$. For $p = 5$: $(p+1)/(2p) = 0.6 < 1/\varphi$. For $p \geq 7$: $(p+1)/(2p) \leq 4/7 < 1/\varphi$. □

Among primes $p \geq 3$, only $p = 3$ produces alignment above $1/\varphi$. The bound holds for every coset and every b-smooth m, without any equidistribution hypothesis.

7. Remarks

The role of equidistribution. The equidistribution condition $|\langle b \rangle \cap B_d| = L \cdot n_d/(p-1)$ holds for all primes below 100 in base 10 except $p = 53, 73, 79, 89$. These exceptions arise when the orbit $\langle b \rangle$ is concentrated in a subset of digit bins. Whether equidistribution holds for a given (p, b) is determined by the interaction between the multiplicative structure of $(\mathbb{Z}/p\mathbb{Z})^*$ and the Euclidean partition induced by δ; see [5] for the general theory of character sums and equidistribution in multiplicative groups. A characterization of equidistributed pairs (p, b) would strengthen Corollary 9 into a theorem with explicit domain.

Connection to the program. The three papers together give:

- The golden ratio paper [1]: the golden ratio's minimal polynomial divides the self-referential cubic only at $p = 3$, producing the identity $\tau^4 = 2 - 3\tau$.
- The digit-partitioning paper [2]: the alignment formula $(2m-1)/(pm-1)$ holds for all primes $p \leq b+1$, characterized by injectivity of δ.
- The present paper: the alignment limit extends to all primes, with corrections from the squared bin sizes of δ.

The next layer is the alignment of composite denominators n with multiple prime factors, where the Chinese Remainder Theorem governs the interaction between prime contributions.

The bin-sum formula. The quantity $S(p,b) = \sum n_d^2$ has a natural interpretation: it is the expected number of remainders sharing a digit bin with a uniformly random remainder. Equivalently, it measures the *collision probability* of the digit function. In Fourier-analytic terms, this sum is the zeroth mode of a Dirichlet kernel expansion [4]. For digit-partitioning primes, the collision probability is minimal ($S = p-1$, each remainder alone in its bin). As p grows past $b+1$, collisions increase quadratically with the bin sizes, enriching the alignment limit.

References

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